INTRODUCTION TO BIOINFORMATICS

ASSIGNMENT 1: SNP ANALYSIS, COMPARATIVE GENOMICS, AND PROTEIN STRUCTURE MODELING

OVERVIEW:

In this project, I was assigned one candidate gene; ATP7B. I explored the genetic variation and transcript diversity of the assigned gene using the ClinVar and NCBI databases. The main objective is to understand how single nucleotide polymorphisms (SNPs) in this gene can affect the DNA and protein sequences, to compare coding transcripts using Multiple Sequence Alignment (MSA), phylogenetic analysis, and pairwise protein alignment, and to predict and analyze 3D protein structures using AlphaFold and TM-align for structural comparison.

CANDIDATE GENE:

ATP7B – ATPase copper transporting beta (copper transport; associated with Wilson's disease)

STEP 1: IDENTIFY SNPS AND THEIR PROPERTIES (CLINVAR DATABASE)

1.2 SEARCH FOR SNPS IN CLINVAR

The three chosen variations:

- RefSeq accession numbers: NM_000053.4, NM_001406511.1, NM_001406512.1

1.3. COLLECT AND DESCRIBE SNP INFORMATION

CLINVAR ID: 3067976

- ClinVar Variation ID= 3067976
- dbSNP ID (rsID)= rs749171049
- HGVS notation (coding) = NM_000053.4:c.4103T>C
- Protein change (if any)= NP_001393441.1:p.Leu1368Pro
- Molecular consequence= missense variant
- Clinical significance = Likely pathogenic
- Transcript(s) affected: NM_000053.4

CLINVAR ID: 617987

- ClinVar Variation ID= 617987
- dbSNP ID (rsID)= rs965523214
- HGVS notation (coding) = NM_001406511.1:c.-54-62G>A
- Protein change (if any)= None
- Molecular consequence= intron variant
- Clinical significance = Uncertain significance

- Transcript(s) affected: NM_001406511.1

CLINVAR ID: 538871

- ClinVar Variation ID= 538871
- dbSNP ID (rsID)= rs28362532
- HGVS notation (coding) = NM_001406512.1:c.-143T>C
- Protein change (if any) = none
- Molecular consequence= 5 prime UTR
- Clinical significance = Benign
- Transcript(s) affected: NM_001406512.1

STEP 2: RETRIEVE CDS FASTA FOR SELECTED REFSEQ TRANSCRIPTS (NCBI)

2.1. DOWNLOAD REFSEQ TRANSCRIPT CDS

Homo sapiens ATPase copper transporting beta (ATP7B), transcript variant 6, mRNA

NCBI Reference Sequence: NM_001406511.1

Homo sapiens ATPase copper transporting beta (ATP7B), transcript variant 1, mRNA

NCBI Reference Sequence: NM_000053.4

Homo sapiens ATPase copper transporting beta (ATP7B), transcript variant 7, mRNA

NCBI Reference Sequence: NM_001406512.1

FASTA Graphics

2.3 AUTOMATED RETRIEVAL

Here is the python transcript for retrieving CDS fasta files:

```
from Bio import Entrez
from Bio import SeqIO
import io


Entrez.email = "shirin.malelpour@gmail.com"

def fetch_cds_sequence(refseq_id):
    handle = Entrez.efetch(db="nucleotide", id=refseq_id, rettype="gb", retmode="text")
    record = SeqIO.read(handle, "genbank")
    handle.close()

    # Extract CDS features
    cds_features = [feature for feature in record.features if feature.type == "CDS"]

    if cds_features:
        # take first cds if there is more than one
        cds = cds_features[0]
        cds_sequence = cds.extract(record.seq)

        # Create a record with the CDS sequence
        cds_record = record[cds.location.start:cds.location.end]
        cds_record.id = f"{refseq_id}_CDS"
```

```
        cds_record.description = f"CDS for {record.description}"

        # Save file
        output_file = f"{refseq_id}.fasta"
        SeqIO.write(cds_record, output_file, "fasta")
        print(f"Saved CDS for {refseq_id} to {output_file}")
        return cds_record
    else:
        print(f"No CDS found for {refseq_id}")
        return None
# Usage
fetch_cds_sequence("NM_000053.4")
```

STEP 3: PERFORM MULTIPLE SEQUENCE ALIGNMENT (MSA) AND PHYLOGENETIC ANALYSIS

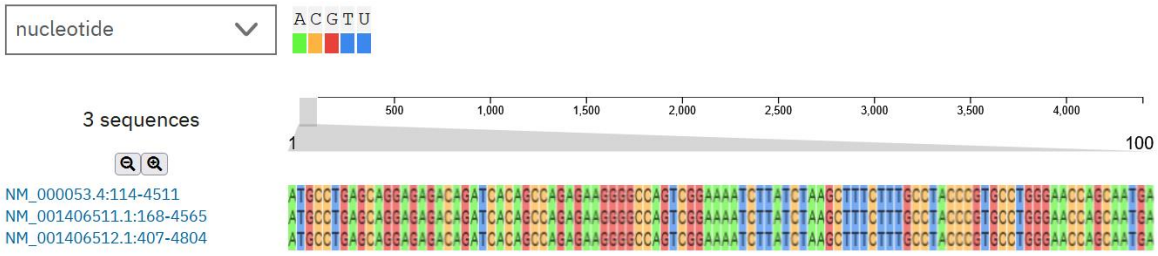
Here is a screen shot of the Clustal multiple sequence alignment results:

```
CLUSTAL O(1.2.4) multiple sequence alignment

NM_000053.4:114-4511        ATGCCTGAGCAGGAGAGACAGATCACAGCCAGAGAAGGGGCCAGTCGGAAAATCTTATCT      60
NM_001406511.1:168-4565     ATGCCTGAGCAGGAGAGACAGATCACAGCCAGAGAAGGGGCCAGTCGGAAAATCTTATCT      60
NM_001406512.1:407-4804     ATGCCTGAGCAGGAGAGACAGATCACAGCCAGAGAAGGGGCCAGTCGGAAAATCTTATCT      60
                            ************************************************************

NM_000053.4:114-4511        AAGCTTTCTTTGCCTACCCGTGCCTGGGAACCAGCAATGAAGAAGAGTTTTGCTTTTGAC      120
NM_001406511.1:168-4565     AAGCTTTCTTTGCCTACCCGTGCCTGGGAACCAGCAATGAAGAAGAGTTTTGCTTTTGAC      120
NM_001406512.1:407-4804     AAGCTTTCTTTGCCTACCCGTGCCTGGGAACCAGCAATGAAGAAGAGTTTTGCTTTTGAC      120
                            ************************************************************

NM_000053.4:114-4511        AATGTTGGCTATGAAGGTGGTCTGGATGGCCTGGGCCCTTCTTCTCAGGTGGCCACCAGC      180
```

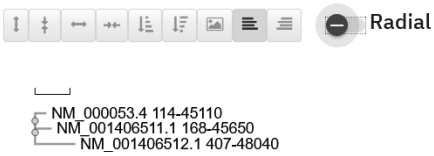


Note: the clustal file of results will be attached to this report.

clustalo-I20251120-083546-0156-61110170-p2m.aln-clustal_num

3.4. CONSTRUCT AND OBTAIN THE PHYLOGENETIC TREE

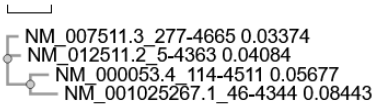
Phylogram



I also tried an MSA with orthologs to this gene in different organisms:

NM_007511.3 (mouse), NM_112511.2 (Rat), NM_000054.3 (Human), NM_001025267.1(Dog)

Here is the result phylogenic tree:



3.5 DISCUSSION

As it is demonstrated, human and dog share the same ancestor, and they both share an ancestor with rat and mouse.

- Which transcript is most similar to the others (based on the tree)?
  - NM_001406511.1(the first graph) and NM_007511.3 (the second graph)
- Which transcript shows the greatest divergence?
  - NM_001025267.1 (dog), NM_001406512.1
- Do SNPs lie in conserved coding regions across the three transcripts?
  - Pay attention to a snippet of the ortholog alignments:
  -

```
NM_007511.3:277-4665        TGGGAGCAATCAATGAAGCAGAGCTTCGCCTTCGACAATGTTGGCTACGAAGGGGGTCTG       177
NM_012511.2:5-4363          TGGGGACAATCAATGAAGCAGAGCTTCGCCTTCGATAATGTTGGCTATGAAGGGGGCCTG       141
NM_000053.4:114-4511        TGGGAACCAGCAATGAAGAAGAGTTTTGCTTTTGACAATGTTGGCTATGAAGGTGGTCTG       144
NM_001025267.1:46-4344      ------------ATGAAGCAGAGCTTTGCCTTTGACAATGTTGGCTACGAAGGTGGCCTG       48
                                         ****** **** ** ** ** ** ********** ***** ** ***
```

  - ATG,AAG,C(A)AG,
  - As you can see the third codon, codes for a different amino acid in the human transcript: CAG= Gln , AAG= lys . This is a **non-synonymous** SNP. The amino acid is *not conserved* (Serine vs. Cysteine). Therefore, this specific SNP is **not in a conserved part** of the coding region.
  - ATG,AAG,C(A)AG,AGC(T)
  - In the second SNP, they both code for the same amino acid(serine), this is a conserved part of the coding region.

3.6 MULTIPLE SEQUENCE ALIGNMENT WITH R

Input and output files will be included. File name: MSA_project folder

```
library(msa)
library(Biostrings)
library(ape)

# Set working directory
setwd("C:/Users/Shirin/OneDrive/Documents/MSA_project")

cat("FILES IN  MSA_PROJECT FOLDER: \n")
files <- list.files()
print(files)
cat("\n")


# STEP 1: LOAD FASTA FILE



fasta_file <- "my_sequences.fasta"


cat("Loading sequences from:", fasta_file, "\n")
sequences <- readDNAStringSet(fasta_file)

# Display sequence information
cat("\n SEQUENCE INFORMATION \n")
cat("Number of sequences:", length(sequences), "\n")
cat("Sequence names:", names(sequences), "\n")
cat("Sequence lengths:", width(sequences), "\n")
cat("\n")
# STEP 2: MULTIPLE SEQUENCE ALIGNMENT

alignment <- msa(sequences, method = "Muscle")

cat("Alignment completed!\n\n")

# STEP 3: VIEW RESULTS

cat("=== MULTIPLE SEQUENCE ALIGNMENT ===\n")
print(alignment)

# Get alignment as a matrix
alignment_matrix <- as.matrix(alignment)

cat("Alignment length:", ncol(alignment_matrix), "\n")
cat("Number of sequences:", nrow(alignment_matrix), "\n")


# STEP 4: SAVE RESULTS

write.phylip(alignment, "alignment_results.phy")

alignment_fasta <- as(alignment, "DNAStringSet")
writeXStringSet(alignment_fasta, "alignment_results.fasta")

capture.output(print(alignment, show="complete"), file="alignment_results.txt")

cat("\n=== RESULTS SAVED ===\n")
cat("1. alignment_results.phy - PHYLIP format\n")
cat("2. alignment_results.fasta - FASTA format\n")
cat("3. alignment_results.txt - Text format\n")



# final file list
cat("\n FINAL FILES IN FOLDER: \n")
print(list.files())
```

STEP 4: PAIRWISE PROTEIN ALIGNMENT (NCBI BLAST)

4.1. OBTAIN PROTEIN SEQUENCES

protein accession ID: NP_001393441.1

subject protein accession ID: NP_001230111.1

Job Title	**ref|NP_001393441.1|**
RID	HY14SJTA114 *Search expires on 11-21 19:07 pm* Download All
Program	Blast 2 sequences Citation
Query ID	NP_001393441.1 (amino acid)
Query Descr	copper-transporting ATPase 2 isoform a [Homo sapiens]
Query Length	1465
Subject ID	NP_001230111.1 (amino acid)
Subject Descr	copper-transporting ATPase 2 isoform c [Homo sapiens]
Subject Length	1354
Other reports	Multiple alignment MSA viewer

.2. PERFORM PAIRWISE ALIGNMENT USING NCBI BLAST

results with BLOSUM62 matrix:

		Max Score	Total Score	Query Cover	E value	Per. Ident	Acc. Len	Accession
			Cluster Representative Sequence					
☑	copper-transporting ATPase 2 isoform c [Homo sapiens]	2270	3042	100%	0.0	86.41%	1354	NP_001230111.1

BLOSUM80:

		Max Score	Total Score	Query Cover	E value	Per. Ident	Acc. Len	Accession
			Cluster Representative Sequence					
☑	copper-transporting ATPase 2 isoform c [Homo sapiens]	2412	3308	100%	0.0	86.41%	1354	NP_001230111.1

BLOSUM90:

		Max Score	Total Score	Query Cover	E value	Per. Ident	Acc. Len	Accession
			Cluster Representative Sequence					
☑	copper-transporting ATPase 2 isoform c [Homo sapiens]	2427	3155	100%	0.0	86.41%	1354	NP_001230111.1

PAM30:

		Max Score	Total Score	Query Cover	E value	Per. Ident	Acc. Len	Accession
			Cluster Representative Sequence					
☑	copper-transporting ATPase 2 isoform c [Homo sapiens]	2422	3148	100%	0.0	84.87%	1354	NP_001230111.1

PAM70:

		Max Score	Total Score	Query Cover	E value	Per. Ident	Acc. Len	Accession
			Cluster Representative Sequence					
☑	copper-transporting ATPase 2 isoform c [Homo sapiens]	2357	3014	100%	0.0	84.72%	1354	NP_001230111.1

4.3. EVALUATE SCORES AND SELECT BEST MATRIX

Best max score: BLOSUM90; which makes sense because the protein sequences are very similar and close to each other.

Alignments:

Score	Expect	Method	Identities	Positives	Gaps
2427 bits(5792)	0.0	Compositional matrix adjust.	1151/1332(86%)	1186/1332(89%)	38/1332(2%)

```
Query  140   AQEAVVKLRVEGMTCQSCVSSIEGKVRKLQGVVRVKVSLSNQEAVITYQPYLIQPEDLRD  199
             +Q A    R+ GMTCQSCV SIE ++  L+G++  KVSL    A + Y P  +  +
Sbjct  55    SQVATSTVRILGMTCQSCVKSIEDRISNLKGIISMKVSLEQGSATVKYVPSVVCLQQVCH  114

Query  200   HVNDMGFEAAIKSKVAPLSLGPIDIERLQSTNPKRPLSSANQNFNNSETLGHQGSHVVTL  259
             ++ DMGFEA+I              E   ++ P R L +
Sbjct  115   QIGDMGFEASIA------------EGKAASWPSRSLPAQE---------------AVV  145

Query  260   QLRIDGMHCKSCVLNIEENIGQLLGVQSIQVSLENKTAQVKYDPSCTSPVALQRAIEALP  319
             +LR++GM C+SCV  IE  + +L GV  ++VSL N+ A + Y P    P  L+  +  +
Sbjct  146   KLRVEGMTCQSCVSSIEGKVRKLQGVVRVKVSLSNQEAVITYQPYLIQPEDLRDHVNDM-  204

Query  320   PGNFKVSLPDGAEG---SGTDHRSSSSHSPGSPPRNQVQGTCSTTLIAIAGMTCASCVHS  376
                F  ++            D     S  P  P     Q    +  +   G    +
Sbjct  205   --GFEAAIKSKVAPLSLGPIDIERLQSTNPKRPLSSANQNFNNSETLGHQGSHVVTLQLR  262
```

- "+" sign means an amino acid with similar biochemical identity
- **High Conservation:** You can see long stretches where the middle line is filled with letters and + signs, confirming the high "Identities" and "Positives" scores from the table.
- **Specific Domain Match:** The sequence GMTCQSCV appears multiple times in both sequences (e.g., Query 144 / Sbjct 59 and Query 268 / Sbjct 148). This strongly suggests this is a conserved **protein domain** or motif that is critical to the function of this protein. It could be a binding site, a structural element, or an enzymatic core.

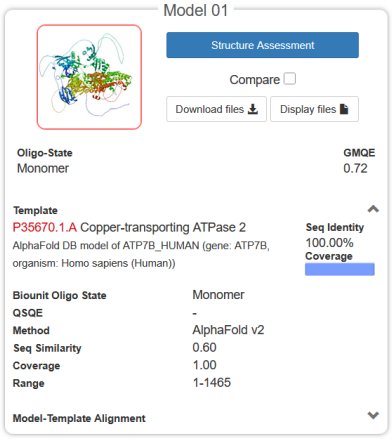
Seqdump(3) seqdump(3).txt

STEP 5: 3D PROTEIN STRUCTURE PREDICTION AND STRUCTURAL COMPARISON (ALPHAFOLD & TM-ALIGN)

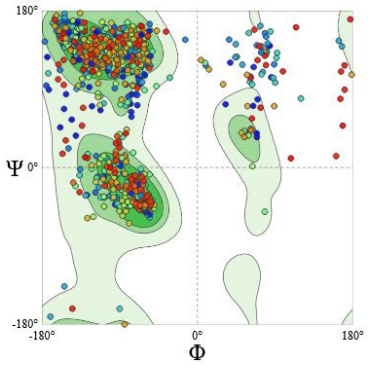
5.1. PREDICT 3D STRUCTURES USING ALPHAFOLD

Alphafold modeling was unavailable. I used Swiss model instead. It is necessary to note that alphafold uses purely computational methods to predict the protein structure whereas Swiss model uses templates from real data for prediction.

Here are the results:







Models

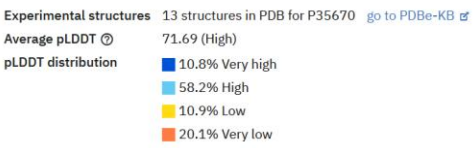
The following model was built (see Materials and Methods "Model Building"):

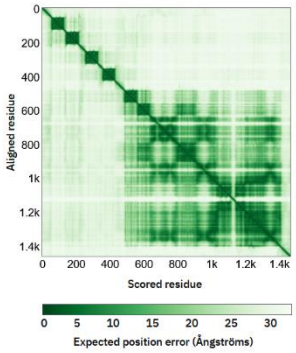
Model #01	File	Built with	Oligo-State	Ligands	GMQE
	PDB	ProMod3 3.6.0	monomer	None	0.72

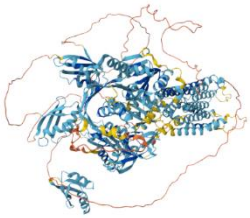
Template	Seq Identity	Oligo-state	QSQE	Found by	Method	Resolution	Seq Similarity	Range	Coverage	Description
P35670.1.A	100.00	monomer	-	AFDB search	AlphaFold v2	-	0.60	1 - 1465	1.00	Copper-transporting ATPase 2

file name : model_01.pdb

I also searched the sequence on Alphafold and found a 100% match with P35670:







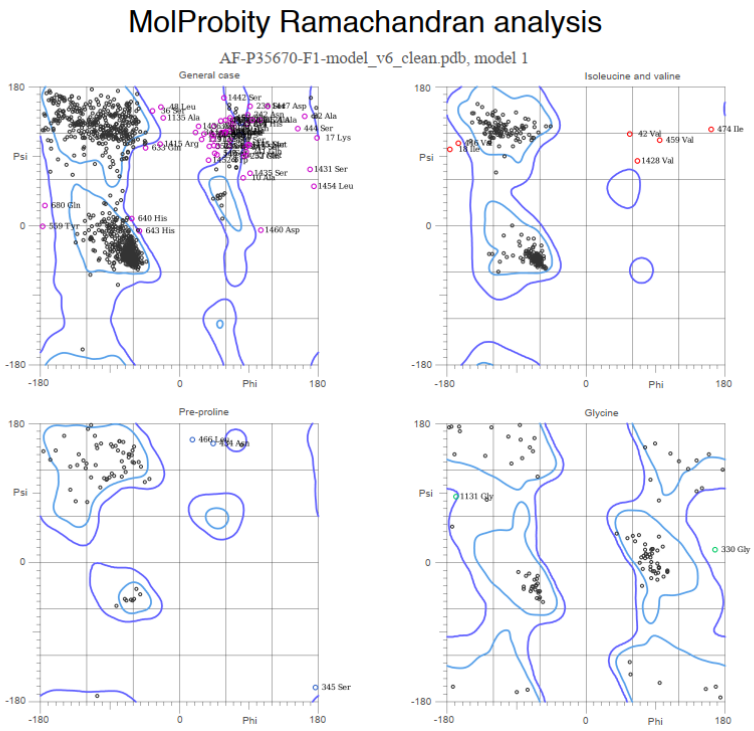
file name: AF-P35670-F1-model_v6.pdb

I also uploaded the above pdb file to MolProbity, here is the favored/nonfavored analysis:

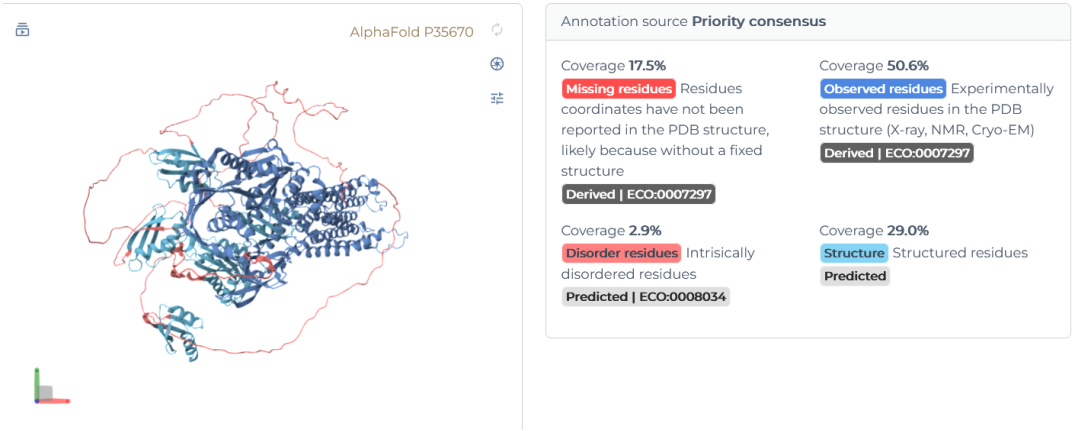
Protein Geometry	Poor rotamers	43	3.51%
	Favored rotamers	1136	92.73%
	Ramachandran outliers	87	5.95%
	Ramachandran favored	1282	87.63%
	Rama distribution Z-score	-1.80 ± 0.20	
	Cβ deviations >0.25Å	27	1.99%
	Bad bonds:	0 / 11203	0.00%
	Bad angles:	186 / 15214	1.22%

And here is the Ramachandran plot:

file name: AF-P35670-F1-model_v6-rama.pdf

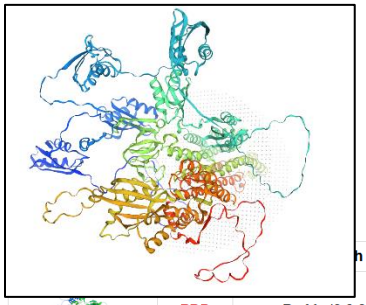


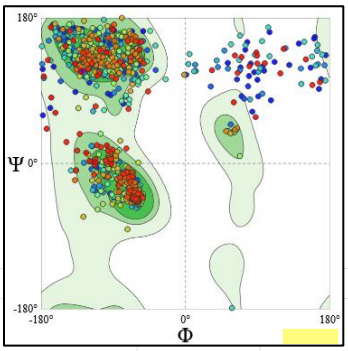
Now here is the Consensus disorder prediction:



MUS MUSCULUS ATPASE, COPPER TRANSPORTING, BETA POLYPEPTIDE (ATP7B), TRANSCRIPT VARIANT 2

- SWISS MODEL:





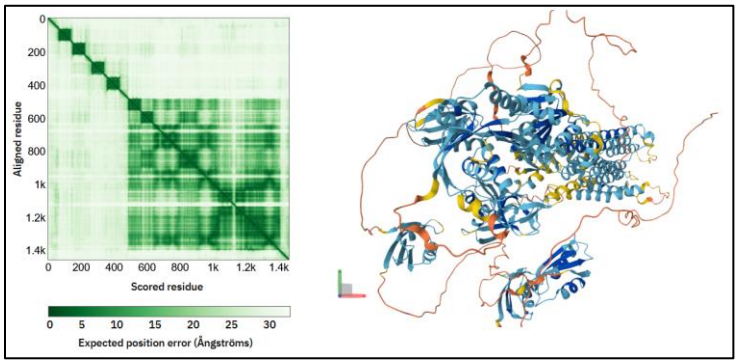
MolProbity Score	1.50
Clash Score	0.45
Ramachandran Favoured	88.42%
Ramachandran Outliers	6.71%

PDB ProMod3 3.6.0

Template	Seq Identity	Oligo-state	QSQE	Found by	Method	Resolution	Seq Similarity	Range	Coverage	Description
Q64446.1.A	100.00	monomer	-	AFDB search	AlphaFold v2	-	0.60	1 - 1462	1.00	Copper-transporting ATPase 2

Pdb file name : model_01mm.pdb

- ALPHA FOLD:

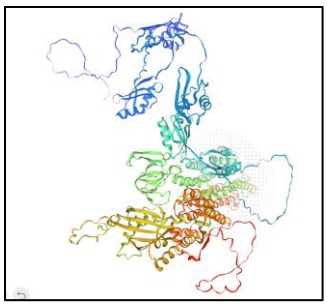


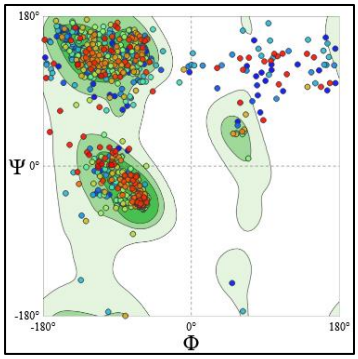
Average pLDDT	71.5 (High)
pLDDT distribution	10.4% Very high
	58.2% High
	11.2% Low
	20.2% Very low

Pdb file name: AF-Q64446-F1-model_v6.pdb

COPPER-TRANSPORTING ATPASE 2 ISOFORM C [HOMO SAPIENS]

SWISS MODEL:





MolProbity Score	1.47
Clash Score	0.73
(A1272 ARG-A1354 ILE), (A705 ARG-A707 GLU), (A148 ARG-A150 GLU)	
Ramachandran Favoured	89.20%
Ramachandran Outliers	6.14%

Model #01	File	Built with	Oligo-State	Ligands	GMQE
	PDB	ProMod3 3.6.0	monomer	None	0.72

Template	Seq Identity	Oligo-state	QSQE	Found by	Method	Resolution	Seq Similarity	Range	Coverage	Description
Q64446.1.A	85.08	monomer	-	AFDB search	AlphaFold v2	-	0.56	1 - 1354	0.99	Copper-transporting ATPase 2

model_01isoform c.pdb

5.2. ANALYZE PREDICTED STRUCTURES

Here is a summary of all the required information:

```
Name: Copper-transporting ATPase 2 isoform A [Homo sapiens]

### Confidence Metrics (pLDDT)
- **Mean pLDDT**: 71.69
- **High-confidence residues (pLDDT > 90)**: 10.8%
- **Core regions (pLDDT > 70)**: 69 % of structure
- **Disordered regions (pLDDT < 70)**: 40 % of structure

### Ramachandran Analysis
- **Favored regions**: 87.63%
- **Outlier regions**: 5.95%
```

```
Name: Mus musculus ATPase, copper transporting, beta polypeptide (Atp7b), transcript
variant 2

### Confidence Metrics (pLDDT)
- **Mean pLDDT**: 71.5
- **High-confidence residues (pLDDT > 90)**: 10.4%
- **Core regions (pLDDT > 70)**: 69.4 % of structure
- **Disordered regions (pLDDT < 70)**: 31.4 % of structure

### Ramachandran Analysis
- **Favored regions**: 88.42%
- **Outlier regions**: 6.71%
```

```
Name: copper-transporting ATPase 2 isoform c [Homo sapiens]

### Confidence Metrics (pLDDT)
```

```
- **Mean pLDDT**: 72
- **High-confidence residues (pLDDT > 90)**: 10 %
- **Core regions (pLDDT > 70)**: 69.1 % of structure
- **Disordered regions (pLDDT < 70)**: 30.4 % of structure

### Ramachandran Analysis
- **Favored regions**: 89.20%
- **Outlier regions**: 6.14%
```

5.3. PERFORM STRUCTURAL ALIGNMENT USING TM-ALIGN

TM-align was unavailable to me, so I used an alternative tool: https://www.ebi.ac.uk/msd-srv/ssm/ssmstart.html

Aligning isoform A Copper-transporting ATPase 2(Homo sapiens) with copper-transporting ATPase 2 isoform c [Homo sapiens]

Query model_01isoform c.pdb:A				Alignment (1 of 1)				Target model_01.pdb:A			
N_{res}	$\%_{res}$	N_{SSE}	$\%_{SSE}$	Q	P	RMSD	N_{align}	N_{res}	$\%_{res}$	N_{SSE}	$\%_{SSE}$
1354	69	77	75	0.409	86.44	0.789	931	1465	64	83	70
SWISS-MODEL SERVER (https://swissmodel.expasy.org) Untitled Project				$\%_{seq}$	Z	N_{SSE}	N_{gaps}	SWISS-MODEL SERVER (https://swissmodel.expasy.org) atp7			
				98.5	35.50	58	9				

```
## Structural Alignment: Isoform C vs Isoform A

### TM-align Equivalent Metrics:
- **TM-score (Q-score)**: 0.409
- **RMSD**: 0.789 Å
- **Aligned length**: 931 residues
- **Sequence identity**: 98.5%

### Interpretation:
- **Fold similarity**: Moderate structural similarity (Q-score = 0.409)
- **Alignment quality**: Excellent backbone superposition (RMSD < 1.0Å)
- **Sequence relationship**: Nearly identical sequences (98.5% identity)
- **Statistical significance**: Highly significant match (Z-score = 86.44)

### Conclusion:
The two isoforms share nearly identical sequences and core structures, with minor
structural variations likely in flexible regions or terminal domains.
```

THEN I ALIGNED ISOFORM C WITH THE ORTHOLOG FROM MUS MUSCULUS:

Query model_01isoform c.pdb:A				Alignment (1 of 1)				Target model_01mm.pdb:A			
N_{res}	$\%_{res}$	N_{SSE}	$\%_{SSE}$	Q	P	RMSD	N_{align}	N_{res}	$\%_{res}$	N_{SSE}	$\%_{SSE}$
1354	80	77	79	0.588	286.17	0.086	1079	1462	74	84	73
SWISS-MODEL SERVER (https://swissmodel.expasy.org) Untitled Project				$\%_{seq}$	Z	N_{SSE}	N_{gaps}	SWISS-MODEL SERVER (https://swissmodel.expasy.org) Untitled Project			
				86.6	53.09	61	2				

```
## Human Isoform C vs Mouse Ortholog

### Structural Alignment Metrics:
- **Q-score**: 0.588
- **RMSD**: 0.086 Å
- **Aligned length**: 1079 residues
- **Sequence identity**: 86.6%
- **Z-score**: 286.17

### Evolutionary Interpretation:
- **High structural conservation**: Q-score > 0.5 confirms same fold across species
- **Excellent backbone alignment**: RMSD < 0.1Å indicates nearly identical core structures
- **Strong sequence conservation**: 86.6% identity reflects functional constraint
- **Statistical significance**: Extremely high Z-score (286.17) validates the match
```

Superposed image: www.ebi.ac.uk/msd-srv/ssm/cgi-bin/ssmserver?qs_jmol=263-85-PH8=921=518=view12_res=view_all

File name: send.rasmol

Isoform A vs mouse

Query model_01.pdb:A				Alignment (1 of 1)				Target model_01mm.pdb:A			
N_{res}	$\%_{res}$	N_{SSE}	$\%_{SSE}$	Q	P	RMSD	N_{algn}	N_{res}	$\%_{res}$	N_{SSE}	$\%_{SSE}$
1465	63	83	69	0.376	74.04	0.823	930	1462	64	84	68
SWISS-MODEL SERVER (https://swissmodel.expasy.org) atp7				$\%_{seq}$	Z	N_{SSE}	N_{gaps}	SWISS-MODEL SERVER (https://swissmodel.expasy.org) Untitled Project			
				90.2	35.32	57	10				

```
### Human Isoform A vs Mouse Ortholog:
- **Q-score**: 0.376
- **RMSD**: 0.823 Å
- **Aligned length**: 930 residues
- **Sequence identity**: 90.2%

### Human Isoform C vs Mouse Ortholog:
- **Q-score**: 0.588
- **RMSD**: 0.086 Å
- **Aligned length**: 1079 residues
- **Sequence identity**: 86.6%
```

Visualization: file name: send(1).rasmol

Alphafold vs Alphafold:

Query AF-P35670-F1-model_v6.pdb:A				Alignment (1 of 1)				Target AF-Q64446-F1-model_v6.pdb:A			
N_{res}	$\%_{res}$	N_{SSE}	$\%_{SSE}$	Q	P	RMSD	N_{align}	N_{res}	$\%_{res}$	N_{SSE}	$\%_{SSE}$
1465	66	84	69	0.362	76.34	1.316	961	1462	66	84	69
ALPHAFOLD MONOMER V2.0 PREDICTION FOR COPPER-TRANSPORTING ATPASE 2 (P35670)				$\%_{seq}$	Z	N_{SSE}	N_{gaps}	ALPHAFOLD MONOMER V2.0 PREDICTION FOR COPPER-TRANSPORTING ATPASE 2 (Q64446)			
				88.0	32.37	58	24				

```
## Method Comparison: SWISS-MODEL vs AlphaFold Predictions

### AlphaFold Human vs Mouse:
- **Q-score**: 0.362
- **RMSD**: 1.316 Å
- **Aligned length**: 961 residues
- **Sequence identity**: 88.0%
```

AlphaFold Predictions Show Poorer Conservation

- **Lower Q-score** (0.362 vs 0.588 for Isoform C)
- **Higher RMSD** (1.316 Å vs 0.086 Å for Isoform C)
- **Fewer aligned residues** than Isoform C comparison

The comparison between SWISS-MODEL (template-based) and AlphaFold (AI-predicted) structures demonstrated that template-based modeling may provide more accurate evolutionary conservation estimates for this specific protein. The exceptional conservation observed in Isoform C using SWISS-MODEL was not replicated in AlphaFold predictions.

Some insights:

- Isoform C shows exceptional conservation with mouse (Q-score = 0.588, RMSD = 0.086Å)
- Isoform A shows significant divergence from mouse (Q-score = 0.376, RMSD = 0.823Å)
- Despite higher sequence identity (90.2% vs 86.6%), Isoform A has poorer structural conservation
- 149 additional residues are structurally conserved in Isoform C compared to Isoform A

Evolutionary Significance:

- Isoform C likely represents the ancestral, evolutionarily constrained form, while Isoform A may have undergone human-specific structural adaptations.

CONCLUSION:

In the first part of this project using ClinVar data base I provided three single nucleotide variations and their clinical significance for ATP7B gene: a likely pathogenic missense variant (p.Leu1368Pro), an intron variant of uncertain significance, and a benign 5' UTR variant.

Furthermore, using biopython tools I retrieved the CDS fasta files and performed multiple sequence alignment on them. What I found was the close and almost identical relationship between the coding sequences of these variants. Then I tried the same analysis with orthologs from different organisms and I found a very interesting

evolutionary relationship: The phylogenetic analysis demonstrated that human and dog share more recent common ancestry compared to rodents.

I performed BLASTP on my original variants that resulted in a peptide sequence variation. I found that blosum90 gave the best maximum score.

And lastly, I did structural analysis between isoform A & C of the human ATP7B protein and a Mouse variant of that same protein. First, I used Swiss model to predict the most likely protein structure for these amino acid sequences. (I also found the predicted structure of the mouse variant and isoform A on Alphafold). All predicted structures showed moderate confidence (mean pLDDT ~71-72) with reasonable Ramachandran statistics (~88% favored regions). Then I aligned these structures and found human isoforms share nearly identical core structures (98.5% sequence identity, RMSD = 0.789Å), yet exhibit moderate Q-scores (0.409), indicating potential functional specializations in flexible regions or termini. Aligning human isoforms with mouse variant I found Isoform C shows exceptional conservation with mouse orthologs (Q-score = 0.588, RMSD = 0.086Å), outperforming Isoform A (Q-score = 0.376).

*I also tried tracking the SNPs from part 1 in the predicted structures and they were mostly in marginal areas with low confidence scores.

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